

Arthur Morris

Email: armo@math.ku.dk

Publications: [Google Scholar](#), [arXiv](#)

Education

King's College, University of Cambridge, Cambridge, England

PhD in Theoretical Condensed Matter Physics

Oct. 2021 - Sept. 2025

- [EPSRC funded](#) PhD in the [Theory of Condensed Matter group](#), supervised by Robert-Jan Slager.
- Researched topological phases of matter, especially multi-band Euler materials.

Corpus Christi College, University of Oxford, Oxford, England

Master's in Mathematical and Theoretical Physics (MMathPhys)

Oct. 2020 - June 2021

- *Distinction* - 77.7%
- BA in Physics
- Part 'B' examinations: *First Class* - 76.1%, 10th/125 in cohort
- Part 'A' examinations: *First Class* - 86.3%, 11th/159 in cohort
- Preliminary examinations: *Distinction* - 82.6% 18th/178 in cohort

Oct. 2017 - Oct. 2020

Publications

7. Constraints on phantom codes from automorphism group bounds *A. S. Morris, D. Malz* [arXiv: 2604.15111](#) (2026)
6. Euler topology in classical spin liquids *L Rüegg, A. S. Morris, H. Yan, R.-J. Slager* [arXiv: 2505.09683](#) (2025)
5. Non-Abelian Hopf-Euler insulators *W. J. Jankowski*, A. S. Morris*, Z. Davoyan, A. Bouhon, F. N. Ünal, R.-J. Slager* (*Contributed equally) [Phys. Rev. B **110**, 075135](#) (2024)
4. Optical manifestations and bounds of topological Euler class *W. J. Jankowski, A. S. Morris, A. Bouhon, F. N. Ünal, R.-J. Slager* [Phys. Rev. B **111**, L081103](#) (2025)
3. Andreev reflection in Euler materials *A. S. Morris, A. Bouhon, R.-J. Slager* [New J. Phys. **26** 023014](#) (2023)
2. Phasik: a Python package to identify system states in partially temporal networks *M. Lucas, A. Townsend-Teague, M. Neri, S. Poetto, A. S. Morris, B. Habermann, L. Tichit* [Journal of Open Source Software, 8\(91\), 5872](#) (2023)
1. Inferring cell cycle phases from a partially temporal network of protein interactions *M. Lucas, A. S. Morris, A. Townsend-Teague, L. Tichit, B. Habermann, A. Barrat* [Cell Reports Methods **3**, 2, 100397](#) (2023)

Experience

[Malz group](#), QMATH, University of Copenhagen Copenhagen, Denmark

Postdoc

Beg. Oct. 1st 2025

- Funded by a [DeiC](#) grant.
- Working on quantum error correction.

[Oka group](#), ISSP, University of Tokyo (Kashiwanoha Campus), Kashiwa, Japan

[JSPS](#) summer scholar

June 17th - Aug. 27th 2024

- Collaborated on projects related to classical spin liquids and the topology of chemical reaction networks.

[Moore group](#), UC Berkeley, Berkeley, California, USA

[Winton-Kavli Cambridge-Berkeley Exchange Scholar](#)

Aug. 18th - Dec. 17th 2023

- Researched the use of Green's functions for the bulk-boundary correspondence in multi-band topological quantum phases.

[NanoDTC](#), Cambridge, UK

[Associate](#)

Dec. 2021 - Present

- Engage with the NanoDTC journal club and events to stay abreast of experimental developments in Energy Materials, Nanoelectronics, Nanophotonics, Sustainable Nano, and Bio-Nanotechnologies.

[Computational Biology \(Habermann\) group](#), CENTURI, Marseilles, France (*completed remotely*)

Summer researcher

30th June - 30th July 2020

- Used Python for gene expression data analysis and clustering to study time-dependent protein interaction networks.

Perm State University, Perm, Russia

[Oxford University Internship Award](#); *Fluid dynamics summer school*

1st July - 10th Aug. 2019

- Courses included fluid dynamics, programming for fluid dynamics, and parallel programming with MPI.
- Used Fortran 90 and 77 to write fluid simulations from scratch (see e.g. [here](#)).

Stanford University, Palo Alto, California, USA

[Kathleen Lavidge Bursary](#)

1st – 21st Sept. 2018

- Took part in a course on ‘The Intersection of Art and Science’, and achieved an A+ in all areas.

[CLASSE REU](#), Wilson Synchrotron Laboratory, Cornell University, Ithaca, New York, USA

[Expanding Horizons Scholar](#)

19th June – 18th July 2018

- Reviewed methods for optimising collimation in accelerator beamlines, published internally as [CBETA technical note 27](#).

CERN, Geneva, Switzerland

24th June – 8th July 2016

- Joined undergraduate summer school students in lectures on accelerator physics, the standard model, and other topics.

Awards and Prizes

- [JSPS](#) summer scholar 2024
- [Winton-Kavli Graduate Exchange Scholar](#) 2023
- [NanoDTC Associate](#) 2021–2024
- [EPSRC PhD Studentship](#) 2021–2024
- [Corpus Christi Academic scholarship](#) for performance in university exams 2018, 19, 20, & 21
- [The Palmer Scholarship for European travel](#) 2021
- [‘A physics prize’](#) from the Oxford Physics department for practical work 2020
- [Corpus Christi travel grant](#) 2018, 19, & 21
- [Oxford University Internship Award](#) 2019
- [Kathleen Lavidge Bursary](#) 2018
- [Expanding Horizons Scholarship](#) 2018

Presentations

- *QLunch*, QMATH, University of Copenhagen, 29/01/25 (1 hr. talk)
- *Seminar*, Quantum Matter Group, Okinawa Institute of Science and Technology, 19/08/24 (1 hr. talk)
- *Seminar*, Yukawa Institute, University of Kyoto, 07/08/24 (1 hr. talk)
- *Seminar*, Institute of Solid State Physics, University of Tokyo, 03/06/24 (1 hr. talk)
- *JSPS Orientation*, SOKENDAI, 19/06/24 (poster)
- *Collective Phenomena Group Meeting*, University of Cambridge, 29/05/24 (1 hr. talk)
- *TI meeting*, University of Pennsylvania, 30/11/23 (1 hr. talk)
- *Quantum Seminar*, UC Berkeley, 06/09/23 (1 hr. talk)

Conferences and summer schools

- QMATH/Qusoft joint outing, Hillerød, Denmark 2nd – 4th June 2026
- DeiC quantum algorithms workshop, Sønderborg, Denmark 20th – 24th May 2026
- Monte Verità Quantum Codes, Ascona, Switzerland 24rd – 30th Apr. 2026
- QIP, Riga, Latvia 24rd – 30th Jan. 2026
- JSPS Orientation, SOKENDAI, Japan 17rd – 20th June 2024
- Condensed Matter in the City, UCL, England 3rd – 7th June 2024
- Condensed Matter in the City, UCL, England 10th – 14th July 2023
- Quantum Dynamics: From Electrons to Qbits, ICTP, Italy 22nd Aug. – 10th Sept. 2022

Teaching

- Supervised 3 final year bachelor student projects at KU Jan. – June 2026
- Supervised 11 (’24) and 13 (’25) Cambridge master’s students in the [Phase transitions](#) course. Feb. – Mar. ’24 & ’25
- Supervised four groups of 2nd year Cambridge undergraduates (total 10 students) in [Mathematical Methods for Natural Sciences](#). One group told me that I was the best supervisor that they had had during their degree. Oct. 2021 – June 2023
- Private tutor in A-Level maths and physics (1 current student, 3 past). One student improved from a D to an A in just two months after I began teaching her. Oct. 2022 – Present

Outreach

- [Runner up](#) in 2023 East-Anglia heats of [IOP Three Minute Wonder](#) competition.
- Was the physics ‘subject ambassador’ for Corpus Christi Oxford, which involved helping with open days and interviews, and promoting a positive image of Oxford to potential applicants and the general public.